

Lab 8

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <WiFiClientSecure.h>

const char* ssid = "Gerry's Fiber";
const char* password = "abc@5678#";

// Use only ONE database URL (your actual project)
const char* URL = "https://exp9-2515c-default-rtdb.asia-southeast1.firebaseio.com/iot.json";

WiFiClientSecure client;

void setup() {
  Serial.begin(115200);

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected!");

  // Required for HTTPS (Firebase)
  client.setInsecure();
}

void loop() {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;

    http.begin(client, URL);
    http.addHeader("Content-Type", "application/json");

    // JSON data to send
    String payload = "{\"value\":\"Internet of Things\"}";

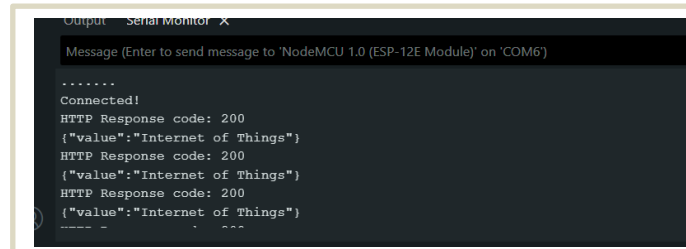
    // PUT = write data
    int responseCode = http.PUT(payload);

    Serial.print("HTTP Response code: ");
    Serial.println(responseCode);

    if (responseCode > 0) {
      String response = http.getString();
      Serial.println(response);
    }

    http.end();

    delay(5000); // send every 5 seconds
  }
}
```



```
Output Serial Monitor x
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM6')
.....
Connected!
HTTP Response code: 200
{"value":"Internet of Things"}
HTTP Response code: 200
{"value":"Internet of Things"}
HTTP Response code: 200
{"value":"Internet of Things"}
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```

